

PROJECT DOCUMENTATION

DOCUMENTATION AND DEMO

EduGenie – Google Gemini Powered Learning Assistant

1. Project Documentation

The project documentation describes EduGenie as a Google Gemini Powered Learning Assistant. It covers the introduction, problem statement, proposed system, objectives, scope, requirements, technology stack, architecture, workflow, modules, implementation, testing, security, future enhancements, outcomes, and conclusion.

2. System Architecture

Student → Web Browser → HTML/CSS Interface → FastAPI Backend → Google Gemini API → Generated Educational Response → FastAPI → Web Browser.

3. Main Modules

Question Answering, Concept Explanation, Quiz Generation, Summarization, and Personalized Learning Path are documented modules. The core implemented module accepts a student question and sends it to Gemini with instructions to return a concise explanation.

4. Demonstration Flow

1. Open the EduGenie webpage.
2. Enter a question or topic.
3. Submit the request.
4. FastAPI receives the request.
5. Backend sends an educational prompt to Gemini.
6. Gemini generates the response.
7. The answer is rendered through Jinja2.
8. The student reads the response.

5. Demonstration Outputs

The project output documentation includes Ask EduGenie and Explain Simply, Generate Quiz and Learning Path, and Summarize Text screens.

6. Demo Result

The documented example test uses the question “Which is the largest ocean?” and records a concise educational answer identifying the Pacific Ocean as the largest ocean.

Prepared for project documentation and academic demonstration.